# Medication, Orders and Treatment Documentation

*Prescriptions, verbal orders, reconciliation, high-alert drugs and transfusion — where documentation and safety are the same thing.*

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|---|---|
| **Module** | 05 of 12 |
| **Study time** | 105 minutes |
| **Audience** | Prescribers, nursing staff, pharmacists, quality and medication safety teams |
| **Assessment** | Knowledge check plus a 50-question scored bank |
| **Standards basis** | NABH 6th Edition MOM chapter; COP chapter on blood transfusion; institutional medication safety policy |

Medication documentation is the only area of the record where a documentation defect reliably produces direct harm within hours. An unclear order is given wrongly; an unsigned administration is given twice; an unreconciled list means a chronic drug stops without anyone deciding to stop it. This module covers the anatomy of a defensible order, verbal orders, high-alert drugs, reconciliation at the three handover points, and the transfusion record chain.

**LEARNING OUTCOMES — by the end of this module you will be able to:**

- Write a medication order containing every required element
- Document allergy status in a way that is visible to everyone who prescribes or administers
- Apply the read-back loop for verbal and telephone orders and complete the countersignature
- Document the additional requirements for high-alert and look-alike sound-alike medications
- Carry out and document medication reconciliation at admission, transfer and discharge
- Complete the transfusion documentation chain so that full traceability exists in both directions
- Document omissions, refusals, adverse drug reactions and medication errors correctly

## 5.1 The anatomy of a medication order

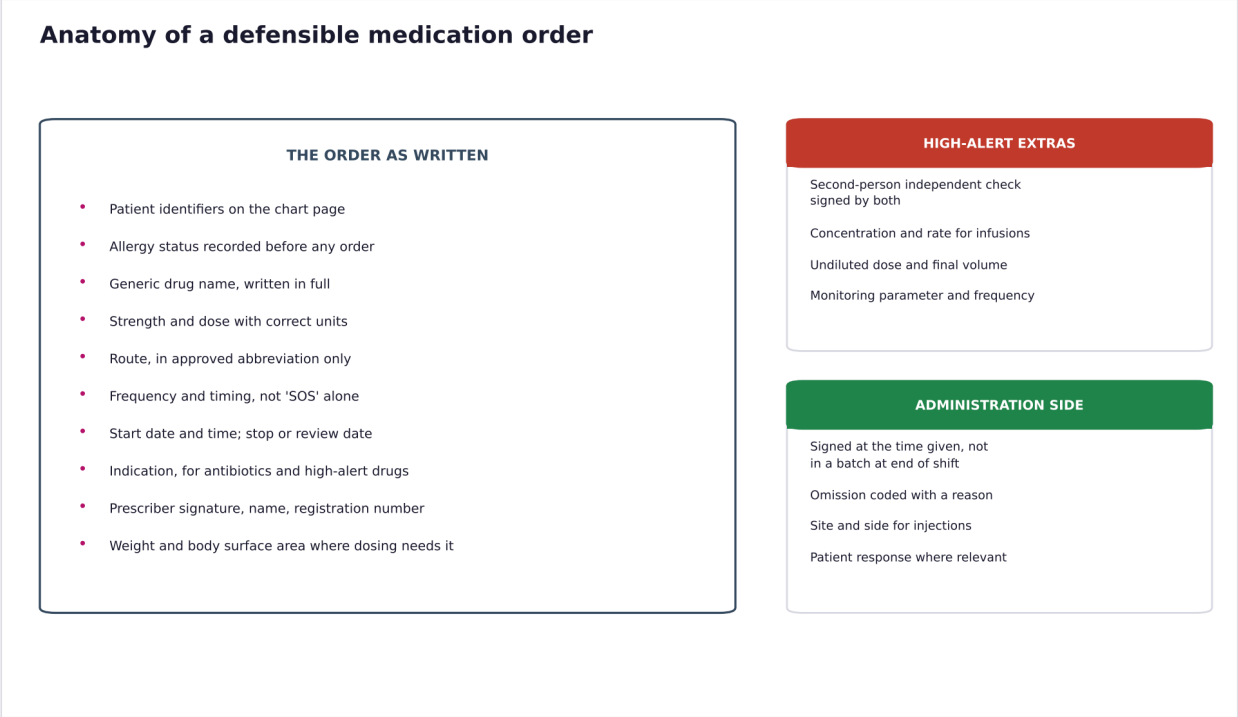



*Figure 5.1 — The elements of a defensible medication order.*

An order that is missing any element is not a minor clerical problem; it is an order that someone must interpret. Interpretation at the point of administration, by a nurse who cannot reach the prescriber, is where medication errors originate.

| Element | Why it is required | What goes wrong without it |
|---|---|---|
| Patient identifiers on the chart page | The chart is a loose document that travels | Doses charted on the wrong patient’s chart |
| Allergy status, completed before any order | Prescribing without it is prescribing blind | A patient receives a drug to which a documented allergy exists elsewhere in the file |
| Generic drug name in full | Brand and abbreviated names are the main source of misreading | Look-alike sound-alike substitution, particularly between similarly named drugs |
| Dose with units written out | Abbreviated units produce ten-fold errors | 4U of insulin read as 40 units |
| Route, in approved form only | Route errors are among the most lethal | Intrathecal administration of a drug intended intravenously |
| Frequency and timing | ‘SOS’ alone sets no maximum and no interval | Repeated dosing beyond the safe limit, particularly with paracetamol and opioids |
| Start date and time | Establishes when therapy began | Duration cannot be established; antibiotic courses run on indefinitely |
| Stop or review date, particularly for antibiotics | Forces a decision rather than drift | Antibiotics continued for weeks with no review, driving resistance and cost |
| Indication for antibiotics and high-alert drugs | Allows review and supports coding and claims | Stewardship review impossible; claim rejected for want of documented indication |
| Prescriber signature, name and registration number | Accountability and the ability to query | Nobody can be asked what was intended |

| Element | Why it is required | What goes wrong without it |
|---|---|---|
| Weight, and body surface area where dosing requires it | Paediatric, chemotherapy and renal dosing depend on it | Dose cannot be checked by the pharmacist or the nurse |

## 5.2 Allergy documentation

Allergy status is the one piece of information that must be visible to every person who prescribes, dispenses or administers, without needing to search. That requires it to be recorded in a fixed location on the chart, on the front of the file, in the electronic record's alert field, and — where hospital policy provides — on a wristband.

- Record the drug or substance, the reaction that occurred, and when it occurred
- Distinguish a true allergy from an intolerance or a side effect, because recording a side effect as an allergy removes a drug class unnecessarily
- Write 'no known allergies' explicitly rather than leaving the field blank; a blank field is indistinguishable from an unasked question
- Update the field if a reaction occurs during the admission, and ensure the update propagates to every location where allergy status appears
- At discharge, ensure the allergy appears on the discharge summary so that the information survives the episode

> **CAUTION**
> An allergy recorded in the admission note but not on the medication chart is, in practical terms, not recorded. The prescriber at 2 a.m. writes on the chart and does not read the admission note.

## 5.3 Verbal and telephone orders

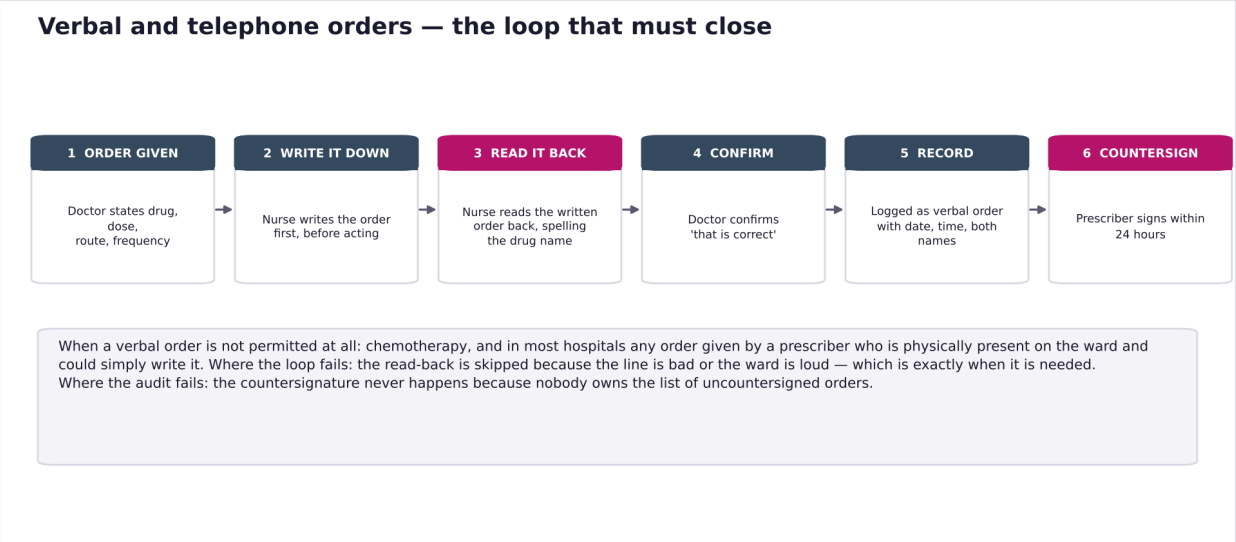



*Figure 5.2 — The verbal order loop, and the point at which it must close.*

Verbal orders exist because clinical work sometimes cannot wait for a prescriber to reach the ward. They are a controlled exception and the controls are precise: the order is written down first, read back in full, confirmed, recorded with both names and the time, and countersigned by the prescriber within the hospital's stated period, conventionally twenty-four hours.

### Read-back in practice

Read-back means repeating the written order back to the prescriber, spelling the drug name where it is similar to another, and stating the dose digit by digit where misunderstanding is possible — "one five milligrams, fifteen". The

prescriber must then confirm. Read-back is skipped most often when the telephone line is poor or the ward is noisy, which is precisely when the risk of mishearing is highest.

| Situation | Is a verbal order acceptable? | What the record must show |
|---|---|---|
| Prescriber physically present on the ward | No — the prescriber should write the order | If exceptionally taken, the reason why writing was not possible |
| Emergency or resuscitation | Yes, with read-back where the situation allows | Drugs, doses, routes and times recorded on the resuscitation record and transcribed to the chart afterwards |
| Telephone order out of hours | Yes, with full read-back | Date, time, prescriber's name, receiver's name, the full order, confirmation of read-back, and countersignature within twenty-four hours |
| Chemotherapy | No, in almost every hospital policy | Written, checked and signed order only; a verbal chemotherapy order is a reportable deviation |
| Change to an existing infusion rate | Only with read-back and a recorded reason | The new rate, the reason, the prescriber, the time, and the countersignature |
| Order received by a person not authorised to take verbal orders | No | The order should be refused and the prescriber asked to contact an authorised person |

**KEY POINT**
Somebody must own the list of uncountersigned verbal orders. In hospitals where no one owns it, the countersignature rate is almost always below fifty per cent, and the hospital has a body of orders that were never authenticated by a prescriber.

## 5.4 High-alert and look-alike sound-alike medications

High-alert medications are those that carry a heightened risk of significant harm when used in error. Hospitals maintain their own list, but it conventionally includes insulin, heparin and other anticoagulants, concentrated electrolytes including potassium chloride, opioids, chemotherapy, neuromuscular blocking agents, and concentrated sedatives.

- Independent double check by a second qualified person, with both signatures and the time, before administration
- The calculation itself recorded where a calculation was required, not merely the final dose
- Concentration and rate recorded for every infusion, with the diluent and final volume
- The monitoring parameter and its frequency documented alongside the order
- For look-alike sound-alike pairs, the drug name written in full and, where the hospital uses it, tall-man lettering applied consistently
- For chemotherapy, the protocol name, cycle number, cumulative dose, body surface area, and the pre-administration parameters that were checked

**CAUTION**
A double check that is signed by two people who looked at the same chart together is not an independent check. The method is that each person independently calculates and verifies, and then they compare. A double check performed as a countersignature adds a signature and no safety.

## 5.5 Administration, omission and refusal

The administration record is the evidence that a prescribed drug reached the patient. Three disciplines carry most of the risk.

1. **Sign at the time of administration, not at the end of the shift.** Batch signing creates a chart that cannot be relied upon for timing, and it is the mechanism by which doses are double-given.
2. **Never sign in advance.** Signing before administration is a false record and, if the dose is then not given, an entirely undetectable omission.
3. **Code every omission with a reason.** A blank box is ambiguous between ‘not given’ and ‘given but not signed’, and the two have opposite clinical implications.

| Code | Meaning | What else must be documented |
|---|---|---|
| A | Patient absent from the ward | Where the patient was, and whether the dose was given later or omitted |
| R | Patient refused | What was explained about the consequences, and whether the prescriber was informed |
| N | Drug not available | The time pharmacy was contacted, and the time the dose was eventually given |
| C | Clinically withheld | The parameter that caused it to be withheld, for example a blood pressure below the stated threshold, and who made the decision |
| F | Patient fasting or nil by mouth | Whether an alternative route was considered and the prescriber informed |
| V | Vomiting or unable to take orally | Whether the dose was repeated or an alternative route used |

A refusal is a decision by a patient with capacity, and it is documented like any other refusal: what was explained, what the risks of not taking the drug are, that the patient understood, and who was informed. Repeated refusals should prompt a clinical review, because they often signal a side effect the patient has not reported.

## 5.6 Medication reconciliation

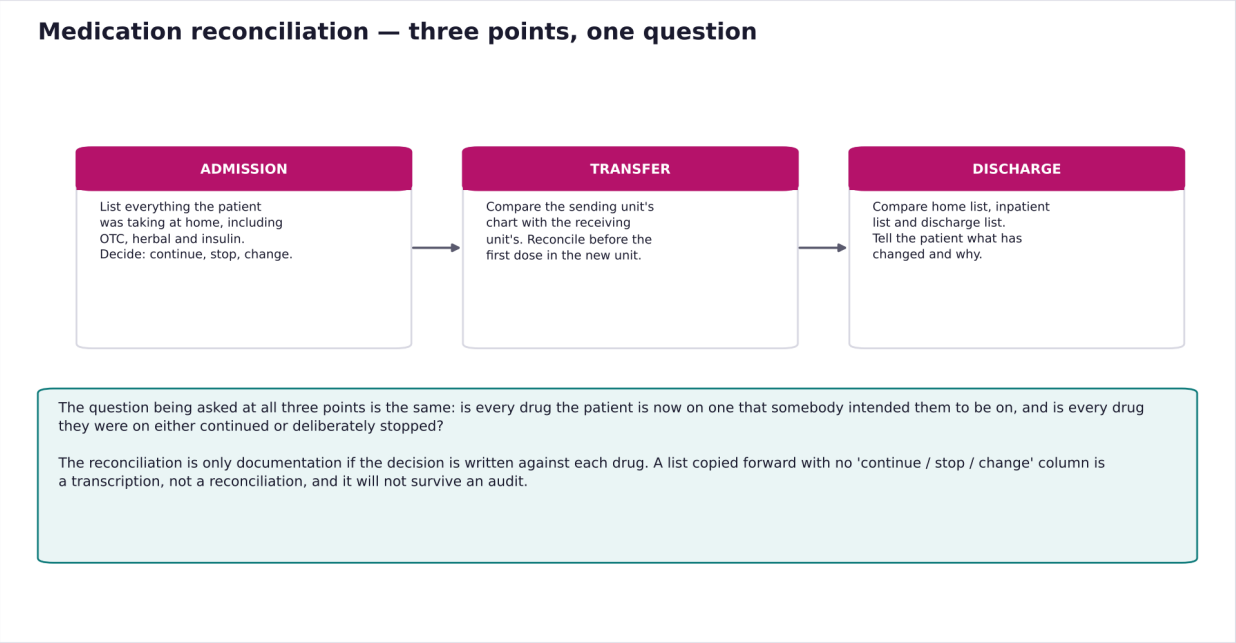



*Figure 5.3 — Reconciliation at the three points where medication lists collide.*

Reconciliation answers one question at each of three points: is every drug the patient is now on a drug somebody intended them to be on, and has every drug they were previously on been either continued or deliberately stopped? Doing this requires a decision recorded against each drug. A list transcribed from one chart to another without those decisions is a transcription, not a reconciliation.

| Point | What to compare | What must be recorded |
|---|---|---|
| Admission | The patient's actual home medication against what is being prescribed in hospital | A complete home list with doses and frequencies, the source of that list, and a continue, stop or change decision against each item with the reason for stopping |
| Transfer between units | The sending unit's chart against the receiving unit's | Reconciliation completed before the first dose in the new unit, with the person who did it identified |
| Discharge | Home list, inpatient list and discharge prescription | What the patient should now take, what has been stopped and why, what is new, what is temporary with a stop date, and evidence that this was explained |

The highest-risk drugs in reconciliation are the ones patients do not mention: inhalers, eye drops, insulin, warfarin, contraceptives, over-the-counter analgesics, and traditional or herbal preparations. Asking specifically about each category is part of the documented process, not an optional extra.

> **KEY POINT**
> The discharge reconciliation is the one patients feel. A patient discharged on a list that silently omits their chronic antihypertensive, or that duplicates it under a brand name alongside the generic, is at direct risk, and the discharge summary is the only control.

## 5.7 Blood and blood products

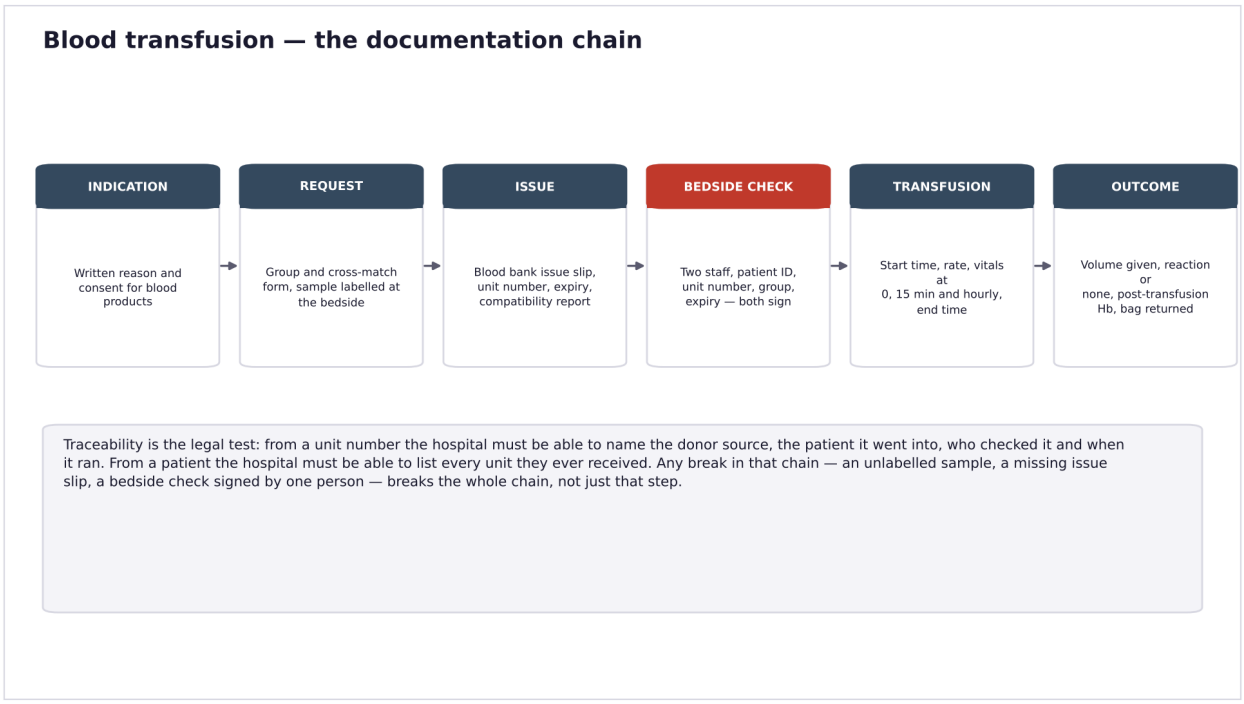



*Figure 5.4 — The transfusion documentation chain.*

Transfusion documentation must support traceability in both directions: from a unit number to the patient who received it, and from a patient to every unit they ever received. Any break in the chain breaks traceability entirely, because the chain is only as good as its weakest link.

1. **Indication and consent.** The clinical reason recorded, and specific consent for blood products obtained and filed.
2. **Sample collection.** The sample labelled at the bedside, by the person who drew it, after positive patient identification — never pre-labelled and never labelled at the nursing station.

3. **Request and compatibility.** The request form with the clinical details, and the compatibility report filed in the record.
4. **Issue.** The blood bank issue record with the unit number, blood group, expiry date and time of issue.
5. **Bedside check.** Two qualified staff independently verifying patient identity, unit number, group, compatibility label and expiry, both signing with the time.
6. **Transfusion.** Start time, rate, observations before starting and at the intervals specified in policy, and the end time.
7. **Outcome.** Volume transfused, reaction or an explicit record of none, post-transfusion investigations, and the disposal or return of the bag.

> **CAUTION**
> The bedside check is the last barrier before an ABO-incompatible transfusion, and it is the step most often compressed. A check signed by one person, or signed by two people of whom one was not present, is a documentation failure that directly precedes the most lethal transfusion error.

## 5.8 Adverse drug reactions and medication errors

An adverse drug reaction is a clinical event and belongs in the clinical record: the drug, the dose, the timing in relation to the dose, the reaction described objectively, the treatment given, the outcome, and the updating of the allergy field. It also belongs in the pharmacovigilance report, which is a separate document.

A medication error is different. The clinical record documents what happened to the patient — what was given, what was done about it, how the patient was monitored, and what the patient was told. The incident report documents the error for the hospital's learning. The two must not be merged: the clinical record should not contain the incident report number as a substitute for clinical detail, and the clinical record should not contain analysis of who was at fault.

| Goes in the clinical record | Goes in the incident report |
|---|---|
| The drug, dose and route actually administered, with the time | The incident classification and severity grading |
| The clinical consequence and the observations taken | The contributing factors and the system analysis |
| The treatment given in response | Names of staff involved, for the hospital's internal process |
| Who was informed, including the prescriber and the family | The corrective and preventive action and its owner |
| What the patient was told about what happened | Trend data and the aggregate reporting |

## 5.9 Controlled substances

Narcotic and psychotropic drugs carry statutory record requirements in addition to clinical ones. The narcotic register must show receipt, issue, administration, wastage and balance, with each entry signed, and the physical count must reconcile to the register at every shift change with two signatures. Wastage must be witnessed and recorded at the time. Discrepancies are reported immediately and are not resolved by adjusting the register.

## 5.10 Infusions and titrated therapy

An infusion that is titrated against a physiological target is a continuously changing prescription, and the record must show both the authority to titrate and what was actually delivered.

- The prescription states the drug, the concentration, the diluent, the final volume, the starting rate, the titration range and the target parameter
- The target is stated in measurable terms — a mean arterial pressure, a sedation score, a blood glucose range — not 'titrate to effect'

- Each rate change is recorded with the new rate, the reason, the time and the person who made it
- The observation that justified the change is recorded alongside it
- The cumulative dose or total volume is calculated per shift
- Line changes, syringe changes and the preparation of each new infusion are recorded with the preparer identified

For insulin infusions and sliding scales, the record must show the blood glucose value, the time it was measured, the dose given against the scale in force, and the next measurement time. A sliding scale running without recorded glucose values is an uncontrolled infusion of a high-alert drug.

## 5.11 Chemotherapy documentation

Chemotherapy carries the strictest documentation requirements of any routine therapy, because the margin between a therapeutic and a lethal dose is narrow and the errors are often irreversible.

| Element | What the record must show |
|---|---|
| Protocol | The named regimen, the cycle number and the day within the cycle |
| Dose calculation | Height, weight, body surface area, the calculated dose, any dose reduction with its reason, and the calculation itself rather than only the result |
| Cumulative dose | Running totals for agents with lifetime limits, checked before each cycle |
| Pre-treatment checks | Blood counts, renal and hepatic function, cardiac assessment where required, and the explicit decision that the patient is fit for this cycle |
| Verification | Independent check by a second qualified person, both signatures with the time; verbal orders are not permitted |
| Administration | Drug, dose, route, sequence, start and end times, the line used, and flushes |
| Monitoring | Observations during administration, extravasation checks for vesicants, and the management of any reaction |
| Patient information | What the patient was told about side effects, when to seek help, and the contact given for out-of-hours advice |

## 5.12 Antimicrobial documentation and stewardship

Antimicrobial prescriptions carry documentation requirements that other drugs do not, because stewardship depends on being able to review them.

- The indication, written on the chart and not only in the notes
- Whether the prescription is empirical or directed by a culture result, and the result if directed
- Cultures taken before the first dose, with the time of collection recorded
- A stop or review date at the point of prescribing — the single most effective stewardship control
- The review decision at forty-eight to seventy-two hours: continue, stop, de-escalate or switch to oral, with the reason
- For restricted antimicrobials, the approval obtained, from whom, and when
- The total duration intended, and at discharge, the duration remaining

> **KEY POINT**
> A prescription with no indication and no stop date cannot be reviewed by anybody, which is precisely why it continues. Both fields exist to force a decision that would otherwise be made by default.

## 5.13 The discharge prescription and counselling

The discharge prescription is where reconciliation becomes visible to the patient, and where errors are carried home. It must be legible, complete and explained.

1. Reconcile against the home list and the inpatient list before writing it.
2. Write each drug with its generic name, strength, dose, route, frequency and duration.
3. State explicitly what has been stopped and what is new, because a list alone does not tell the patient that their old tablet should no longer be taken.
4. Where a drug is temporary, give the stop date; where it needs monitoring, say what and when.
5. Counsel the patient and record that counselling, naming who was counselled.
6. Check that the discharge summary and the prescription agree; where they differ, the patient will follow whichever they read.

## 5.14 Medication documentation in special situations

| Situation | Additional documentation |
|---|---|
| Patient’s own medication used in hospital | Identification and verification of the drug, the decision to use it, and the storage arrangement |
| Self-administration by the patient | The assessment that the patient is competent to self-administer, the agreement, what supervision applies, and how administration is recorded |
| Drug given during transport or in an ambulance | A transport record showing drug, dose, route and time, transcribed into the hospital record on arrival |
| Investigational or trial medication | Ethics approval reference, protocol identifier, randomisation or kit number, accountability record, and the separate trial documentation |
| Drug shortage and therapeutic substitution | The substitution made, the authority for it, and that the prescriber and the patient were informed |
| High-risk route such as intrathecal | The specific procedural checks required by policy, the second check signed, and the location of administration |

## 5.15 Putting this module to work

1. Take one medication chart and check it against the element list in Figure 5.1. Count how many elements are missing.
2. Check whether the allergy field on that chart is completed, and whether it matches the admission note.
3. Find out who in your hospital owns the list of uncountersigned verbal orders, and what the current countersignature rate is.
4. Look at the last ten omission boxes on your ward’s charts. How many are blank rather than coded?
5. Take one discharge summary and check whether it states what was stopped and why, not only what to continue.

# Knowledge Check

Work through each item before reading the model answer. These are applied questions: the answer is the standard of documentation your assessor, your medico-legal adviser or the next clinician to read the file would expect.

Q1. **A nurse receives a telephone order for ‘Inj Lasix 40 stat’, gives it, and charts it after administration with the doctor’s name. What is wrong with the sequence and what are the risks?**

**Answer:** The order should have been written down first, read back in full with the drug name spelled and the dose stated digit by digit, confirmed by the prescriber, and only then administered. Charting afterwards loses the read-back, and the order as recorded also lacks the route and the unit of dose. Risks include the wrong drug being given where the name was misheard, and the hospital being unable to show that the order was confirmed. The countersignature within twenty-four hours is also required and is frequently not obtained.

Q2. **A patient receives potassium chloride by infusion. The chart records the drug and the dose but not the concentration, the diluent, the final volume or the rate. Why does this matter beyond documentation?**

**Answer:** Concentrated potassium is a high-alert drug for which the rate and concentration determine whether the infusion is safe or lethal. Without them the nurse at the next shift cannot verify what is running, the pharmacist cannot check the order, and if the patient develops an arrhythmia nobody can establish what was being delivered. The order should specify the dose, the diluent, the final volume, the concentration, the rate, the route and the monitoring required, with an independent double check signed by two people.

Q3. **A patient on long-term warfarin is admitted for pneumonia. The admission chart does not list warfarin and it is not given for six days. What failed, and at which point?**

**Answer:** Medication reconciliation at admission failed. A complete home medication list should have been taken, its source documented, and a continue, stop or change decision recorded against every item with a reason. The failure is compounded if the admission note recorded ‘on regular medication’ without a list. The consequence is an unintended cessation of anticoagulation with no clinician having decided it, which is the classic reconciliation harm.

Q4. **A bedside transfusion check is signed by two nurses, but one of them signed at the nursing station because she was called away. Is this a documentation problem or a safety problem?**

**Answer:** Both, and they are the same problem. The purpose of the two-person check is two independent verifications at the bedside against the patient and the unit. A signature given away from the bedside attests to a check that did not occur, so the record is false and the last barrier against an ABO-incompatible transfusion has been removed. This should be reported as an incident even though no harm occurred.

Q5. **An omission box on a medication chart is blank. The nurse says the dose was given but she forgot to sign. How should this be resolved?**

**Answer:** It cannot be resolved by signing the box now, which would make a false contemporaneous entry. She should make a late entry: sign and date at the current time, annotate it as a late entry for the original time, and write a nursing note recording the drug, dose, route, actual administration time and the reason for the late entry. The underlying issue — batch signing or signing after the round — should be addressed, because a blank box is clinically ambiguous and risks a duplicate dose.

Q6. **A patient develops a rash after a dose of ceftriaxone. What must be recorded, where, and what must change for the rest of the admission?**

**Answer:** In the clinical record: the drug, dose and time, the interval before the reaction, an objective description of the rash and any systemic features, the treatment given, the outcome, and that the prescriber and the patient were informed. The allergy field must be updated everywhere it appears — chart, file front, electronic alert and wristband — and the drug and its class reviewed for the rest of the admission. A pharmacovigilance report is submitted separately, and the allergy must appear on the discharge summary so the information survives the episode.

**ASSESSMENT BANK**

A fuller bank of 50 scored questions for this module is supplied separately as a JSON question bank for loading into your LMS.